IMO(23rd) – 1982

First Day

1. The function $f(n)$ is defined for all positive integers n and takes on non-negative integer values. Also, for all m,n : $f(m+n)-f(m)-f(n)=0$ *or* 1, $f(2)=0$, $f(3)>0$ and $f(9999)=3333$. Determine $f(1982)$.
2. A non-isosceles triangle $A_1A_2A_3$ is given with sides a_1,a_2,a_3 (a_i is the side opposite to A_i). For all $i=1,2,3$, M_i is the mid point of side a_i, and T_i is the point where the incircle touches side a_i. Denote by S_i the reflection of T_i in the interior bisector of angle A_i. Prove that the lines M_1S_1, M_2S_2 and M_3S_3 are concurrent.
3. Consider the infinite sequences $\{x_n\}$ of positive real numbers with the following properties: $x_0=1$ and for all $i\ge 0$, $x_{i+1}\le x_i$.
   (a) Prove that for every such sequence, there is an $n\ge 1$ such that:
   $$\frac{x_0^{\ 2}}{x_1}+\frac{x_1^{\ 2}}{x_2}+\ldots+\frac{x_{n-1}^{\ \ 2}}{x_n}\ge 3.999$$
   (b) Find such a sequence for which $\frac{x_0^{\ 2}}{x_1}+\frac{x_1^{\ 2}}{x_2}+\ldots+\frac{x_{n-1}^{\ \ 2}}{x_n}<4$.

Second Day

4. Prove that if n is a positive integer such that the equation $x^3-3xy^2+y^3=n$ has a solution in integers (x,y), then it has at least three such solutions. Show that the equation has no solution in integers when $n=2891$.
5. The diagonals AC and CE of the regular hexagon ABCDEF are divided by the inner points M and N, respectively, so that $\frac{AM}{AC}=\frac{CN}{CE}=r$. Determine r if B, M and N are collinear.
6. Let S be a square with sides of length 100, and let L be a path within S which does not meet itself and which is composed of line segments $A_0A_1, A_1A_2, \ldots, A_{n-1}A_n$ with $A_n\ne A_0$. Suppose that for every point P the boundary of S there is a point of L at a distance from P not greater than $\frac{1}{2}$. Prove that there are two points X and Y in L such that the distance between X and Y is not greater than 1, and the length of that part of L which lies between X and Y is not smaller than 198.

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